

# Online Payments Fraud Detection System – Project Report

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## 1. Project Title

### Machine Learning Based Online Payments Fraud Detection System

## 2. Project Description

This project implements a machine learning-based fraud detection system designed to identify fraudulent online payment transactions in real-time. The system analyzes transaction details such as amount, time, user behavior, and device information to classify transactions as **Fraudulent** or **Legitimate**.

The system provides an intuitive dashboard for monitoring transactions and integrates a backend API for real-time fraud prediction.

### 2.1 Problem Statement

Online payment fraud is increasing rapidly due to:

- Growth of digital transactions
- Phishing and cyber attacks
- Card-not-present fraud
- Identity theft

Traditional rule-based systems:

- Fail to detect evolving fraud patterns
- Generate high false positives
- Affect genuine customer experience
- This project aims to provide an intelligent, adaptive fraud detection system using machine learning.

### 2.2 Solution

A Flask/FastAPI-based fraud detection system that:

- Accepts transaction inputs via API/UI
- Performs preprocessing and feature engineering
- Uses trained ML models to classify transactions
- Provides fraud probability scores

- Triggers alerts for suspicious activities
- Displays monitoring dashboard for analysts

### 3. Technologies Used

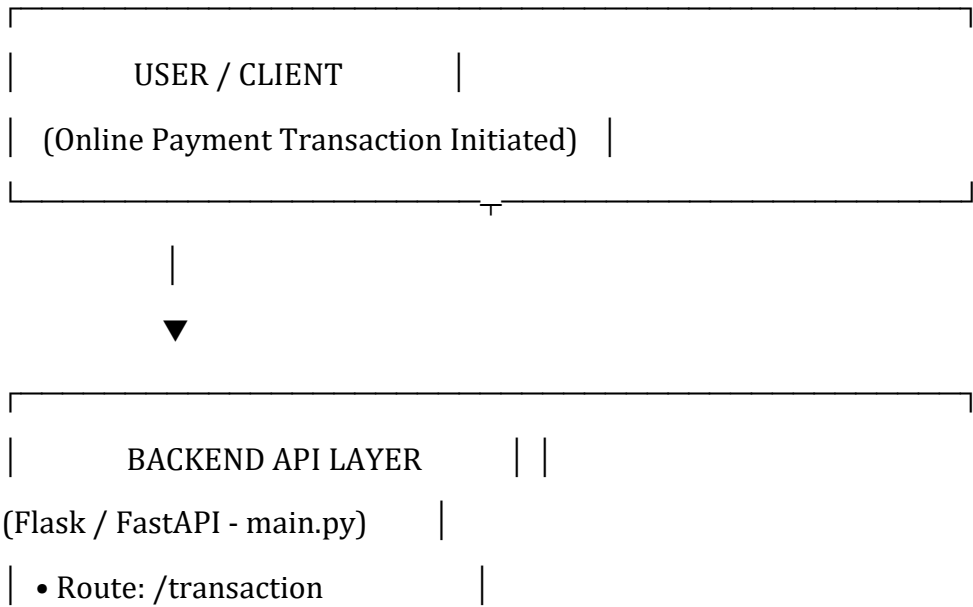
#### 3.1 Backend

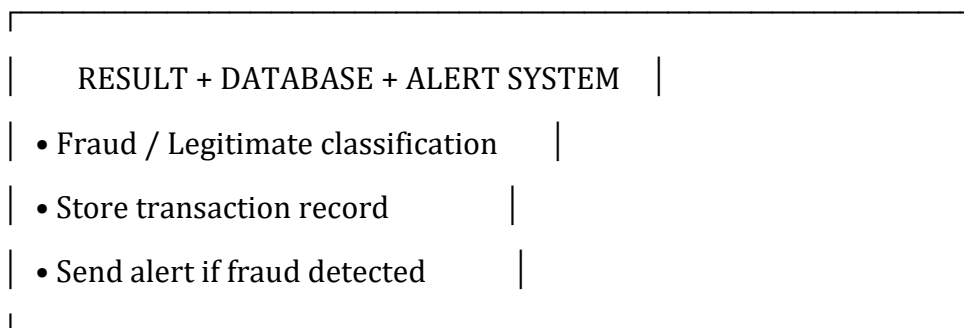
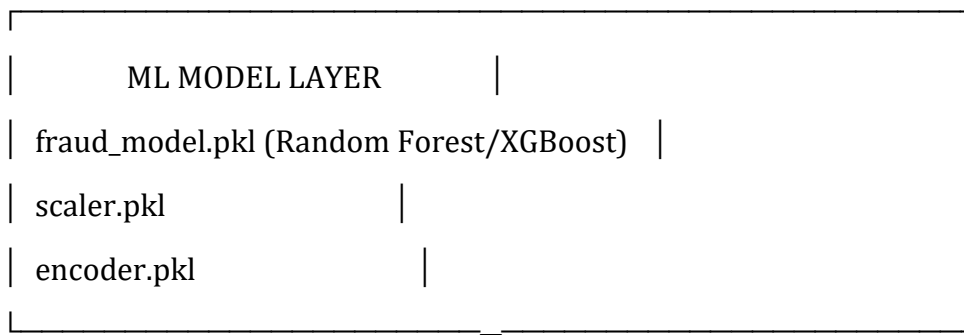
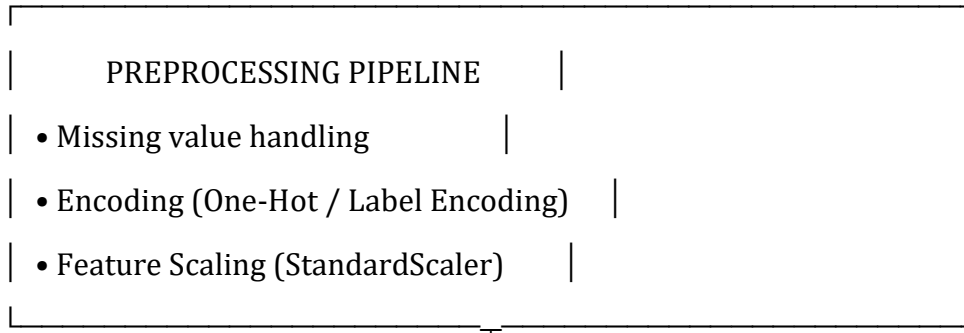
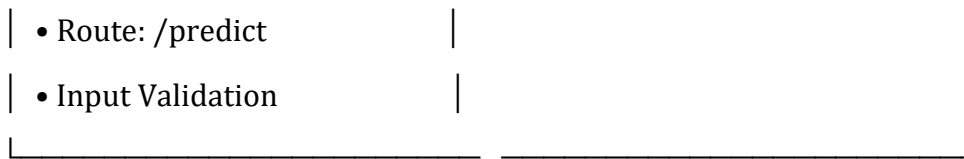
| Technology      | Purpose                   |
|-----------------|---------------------------|
| Python 3.x      | Programming language      |
| Flask / FastAPI | Web framework             |
| Pandas          | Data manipulation         |
| NumPy           | Numerical computations    |
| Scikit-learn    | Machine learning modeling |
| Joblib / Pickle | Model serialization       |

#### 3.2 Frontend

| Technology | Purpose                   |
|------------|---------------------------|
| HTML5      | Page structure            |
| CSS3       | Styling                   |
| JavaScript | Interactivity             |
| Chart.js   | Fraud trend visualization |
| Bootstrap  | Responsive UI             |

### 4. System Architecture





## 5. Features Implemented

### 5.1 Transaction Processing Module

- Accepts transaction details
- Validates input parameters
- Generates fraud probability score
- Stores transaction logs

### 5.2 Fraud Monitoring Dashboard

- Displays flagged transactions
- Shows fraud percentage trends
- Provides daily/weekly fraud statistics

### 5.3 Alert System

- Real-time fraud alerts
- Suspicious transaction notification
- case logging

## 6. Machine Learning Model

### 6.1 Dataset

- Source: Public Credit Card Fraud Dataset (Kaggle)
- Records: 284,807 transactions
- Fraud Cases: 492 (~0.17%)
- Target Variable: Class (0 = Legitimate, 1 = Fraud)

### 6.2 Preprocessing

- 1) Handling imbalanced dataset (SMOTE)
- 2) Feature scaling (StandardScaler)
- 3) Train-Test Split (80-20)
- 4) Feature engineering (transaction frequency, anomaly detection)

### 6.3 Model Training •

Algorithm Used:

- Logistic Regression
- Random Forest
- XGBoost (Best Performing)
- Cross-Validation: 5-Fold

- Hyperparameter Tuning: GridSearchCV

## 6.4 Model Performance

| Metric    | Score |
|-----------|-------|
| Accuracy  | ~99%  |
| Precision | ~92%  |
| Recall    | ~85%  |
| F1-Score  | ~88%  |
| ROC-AUC   | ~0.97 |

## 7. Files Modified/Created

| File                     | Description                      | Type      |
|--------------------------|----------------------------------|-----------|
| main.py                  | Backend API and prediction logic | Created   |
| model_training.ipynb     | Model training notebook          | Created   |
| fraud_model.pkl          | Trained ML model                 | Generated |
| scaler.pkl               | Feature scaler                   | Generated |
| encoder.pkl              | Feature encoder                  | Generated |
| templates/dashboard.html | Monitoring UI                    | Created   |
| static/css/style.css     | Styling                          | Created   |

## 8. How to Run

### 8.1 Prerequisites

- Python >= 3.8
- Flask / FastAPI
- Pandas
- NumPy
- Scikit-learn

### 8.2 Installation

*# Clone repository*

```
git clone <repository-url> cd
Fraud_Detection_System
```

*# Create virtual environment*

```
python -m venv .venv
.venv\Scripts\activate
```

*# Install dependencies* pip install flask  
pandas numpy scikit-learn

```
# Run application
python main.py
```

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## 9. Application Link

**Local Development URL:** <http://localhost:5000>

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## 10. Future Enhancements

1. Deep Learning (Neural Networks)
2. Real-time streaming with Apache Kafka
3. Mobile fraud alert integration
4. Multi-bank fraud intelligence sharing
5. Cloud deployment (AWS/Azure/GCP)

## 11. Conclusion

This project successfully demonstrates the integration of Machine Learning with backend systems to build a real-time fraud detection solution. The application:

- Detects fraudulent transactions accurately
- Reduces financial losses
- Minimizes false positives
- Enhances customer trust in digital payment systems
- Provides scalable and modular architecture

## 12. References

1. Kaggle Credit Card Fraud Detection Dataset
2. Flask Documentation – <https://flask.palletsprojects.com/>
3. Scikit-learn Documentation – <https://scikit-learn.org/>
4. XGBoost Documentation

**Report Prepared By:** NOSSAM PAVAN KUMAR REDDY

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**Course:** Artificial Intelligence and Machine Learning